

Week 1: Assignment Solutions

1. Which of the following statements are true?
 - a. The ENIAC computer was built using mechanical relays.
 - b. Harvard Mark1 computer was built using mechanical relays.
 - c. PASCALINE computer could multiply and divide numbers by repeated addition and subtraction.
 - d. Charles Babbage built his automatic computing engine in 19th century.

Solution: ((b) and (c)) ENIAC was built using vacuum tubes. Charles Babbage designed his automatic computing engine but could not built it.

2. Which of the following statements are true for Moore's law?
 - a. Moore's law predicts that power dissipation will double every 18 months.
 - b. Moore's law predicts that the number of transistors per chip will double every 18 months.
 - c. Moore's law predicts that the speed of VLSI circuits will double every 18 months.
 - d. None of the above.

Solution: (b) Moore's law only predicts that number of transistors per chip will double every 18 months.

3. Which of the following generates the necessary signals required to execute an instruction in a computer?
 - a. Arithmetic and Logic Unit
 - b. Memory Unit
 - c. Control Unit
 - d. Input/Output Unit

Solution: (c) Control unit acts as the nerve center of a computer and generates the necessary control signals required to execute an instruction.

4. An instruction ADD R1, A is stored at memory location 4004H. R1 is a processor register and A is a memory location with address 400CH. Each instruction is 32-bit long. What will be the values of PC, IR and MAR during execution of the instruction?
 - a. 4004H; ADD R1, A; 400CH
 - b. 4008H; ADD; 400CH
 - c. 4008H; ADD R1, A; 400CH
 - d. None of the above

Solution: (c) As each instruction is 32-bit, the PC will get incremented by 4. So PC will be 4008H. Instruction Register (IR) stores the instruction that is fetched from memory; hence it will store ADD R1, A. And Memory Address Register (MAR) will contain either the address of the instruction that is to be fetched from memory or the address of the memory operand. In this

instruction the memory operand is A whose address is 400CH; hence MAR will contain 400CH during the execution of the instruction.

5. The task of the Program Counter (PC) is to
- Point to the next instruction to be executed.
 - Count the total number of instructions in a program.
 - Point to the current instruction that is being executed.
 - None of the above.

Solution: (a) It holds the memory address of the next instruction to be executed.

6. A computer has 2GB (Gigabytes) of byte addressable memory. The number of address lines will be [31](#).

Solution: Since $2\text{ GB} = 2 \cdot 2^{30} = 2^{31}$, the number of address lines required will be 31.

7. Consider a 1 MB (Mega byte) byte addressable memory system, with word size of 32 bits. The number of bits in MAR and MDR will be:
- 23,32
 - 20, 8
 - 23, 32
 - 20, 32

Solution: (d) MAR will require 20 bits (since $1\text{ M} = 2^{20}$) as it stores the address of an instruction or data, and MDR will require 32 bits as it will store either an instruction or data.

8. A hexadecimal number A45DF7BC is stored from memory address 3007H in little endian format. The byte stored in memory locations 3007H, 3008H, 3009H, and 300AH will be:
- BC, F7, 5D, A4
 - A4, 5D, F7, BC
 - BC, F7, A4, 5D
 - F7, BC, A4, 5D

Solution: (a) In Little Endian the least significant byte is stored at lower address followed by most significant byte. Hence first BC will be stored at 3007H, then F7 will be stored at 3008H, and so on.

9. Which of the following statements are false for Harvard architecture?
- The program and data are stored in separate memory units.
 - The program and data are stored in same memory.
 - Processor fetches instructions from program memory and accesses data from data memory.
 - The program memory and data memory can be built using different memory technologies.

Solution: (b) The program and data are not stored in same memory in Harvard architecture; they are stored in separate memories. In contrast, for

von Neumann architecture, program and data are stored in the same memory.

10. Which of the following instructions are valid in stack architectures?

- a. ADD
- b. PUSH X
- c. ADD X
- d. POP

Solution: ((a) and (b)) In stack architecture, we need not have to specify the operands. They are implicitly taken from the top of the stack. Hence ADD X is not valid. Also when we POP we need to specify where to store the data; hence it should be POP X.

11. The minimum number of instructions required to execute the statement $A = (B+C)/D$ using accumulator-based architecture is 4, where A, B, C and D are memory locations.

Solution: The answer is 4, corresponding to the following instructions:

```
LOAD  B
ADD   C
DIV   D
STORE A
```

12. The minimum number of instructions required to execute the statement $A = (B+C)/D$ using load store architecture is 6, where A, B, C and D are memory locations.

Solution: The answer is 6, corresponding to the following instructions:

```
LOAD  R1, B
LOAD  R2, C
ADD   R1, R2
LOAD  R3, D
DIV   R1, R3
STORE R1, A
```